DO CONTEXT ENGINES HELP? RETRIEVAL QUALITY, DETERMINISM, AND AGENT UTILITY UNDER MATCHED BASELINES AND AUDITED EXPOSURE

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ABSTRACT

We ask whether a repository *context engine*—a retrieval stack that indexes a codebase and returns files to a coding agent—improves on what a practitioner would assemble from standard parts, and whether it changes what the agent repairs. Answering this is difficult because published margins are typically measured against weak lexical baselines, on evaluation sets that development had already seen, and across engines whose output contracts differ. We evaluate one open-source engine, Delphi, under a protocol that (i) records for every evaluation case whether any development decision could have seen it, (ii) compares against a ladder of conventional retrievers that add Delphi's own components one at a time to a dense+BM25 hybrid built from the same embedding model, (iii) matches output contracts across two hosted engines by routing every engine's retrieved evidence through one frozen synthesis stage, and (iv) runs a preregistered executable pilot on the same tasks as the ranking evaluation. On 160 SWE-bench Verified localization instances never seen by development, Delphi leads the full conventional stack by $+0.06$–0.08 on Recall@5, Recall@20, and budget-constrained yield (repository-cluster 95% intervals excluding zero; MRR tied), while on an 18-case repository-disjoint set the conventional stack leads; on every set the reranking head accounts for most of the margin over lexical baselines ($+0.3$ MRR versus $+0.06$ for everything else). Under matched contracts the documentation engines are statistically indistinguishable at $n{=}40$, the hosted synthesis engine's retrieval is at least as stable as Delphi's (1.000 vs. 0.980 retrieved-set repeatability), and a Delphi seed changes verified repair by $+0.048$ $[-0.008, +0.113]$ over no seed and by -0.008 over random files (mini-SWE-agent, 62 instances, two trajectories each). We release the code, per-case artifacts, agent trajectories, and the exposure ledger.

1 INTRODUCTION

Coding agents increasingly rely on a context engine: a service that indexes a repository and, for each agent query, returns the files the agent should read. Such engines combine dense retrieval, lexical matching, structural indices, learned rerankers, and query expansion, and report large gains over lexical baselines. Three properties of those evaluations make the gains hard to interpret. The evaluation cases were often available, at least in aggregate, while the engine's components were selected, so a reported margin partly measures selection. The baselines are usually lexical, so a margin cannot be attributed to the engine's design rather than to a learned reranker that would help any candidate pool. Comparisons with hosted engines mix output contracts—raw context against synthesized answers—and measure different quantities under one name, determinism among them. Finally, retrieval quality is rarely connected to what an agent repairs.

We study these questions for one open-source engine, Delphi (hybrid lexical/vector candidates, query expansion, a cross-encoder, a listwise reranker), whose own history exhibits the first prob-

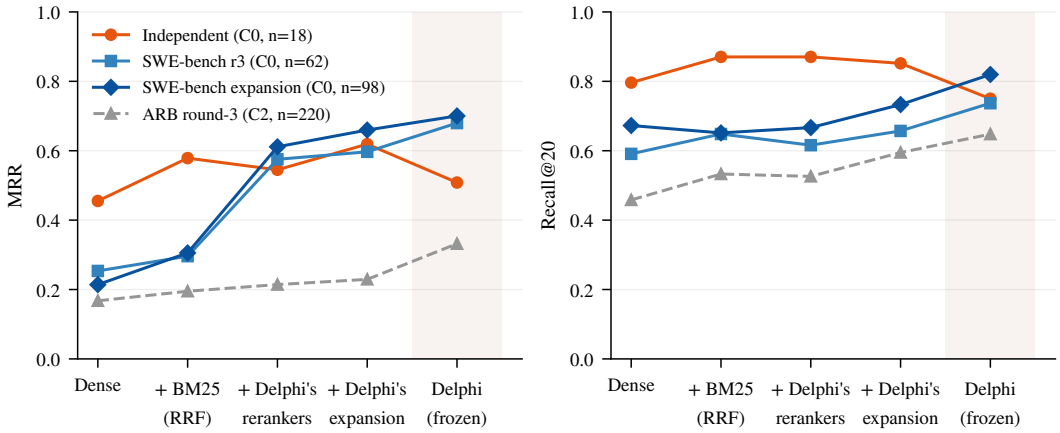

Figure 1: Where the margin comes from. Each rung of the matched ladder adds one Delphi component to the previous rung over the identical corpus (§3); the last point is the frozen Delphi stack. On issue-style queries (SWE-bench Verified) the reranking head moves MRR by roughly $+0.3$ and everything Delphi adds on top by $+0.04$–0.08; on the commit-to-files set Delphi finishes below the full conventional stack. The dashed line is the re-partitioned ARB set on which Delphi's components were selected (exposure class C2) and is shown for context only.

lem: its largest evaluation partition (220 Agent Retrieval Bench cases) was re-drawn from a pool every case of which had been scored in an earlier development round whose aggregates gated which components shipped. We ask three questions under a protocol that removes each confound.

1. **RQ1 (ranking).** Against lexical baselines, a component-matched conventional hybrid, and hosted engines, where does Delphi lead, tie, or trail on cases that no development decision could see?
2. **RQ2 (determinism).** Where does run-to-run variance come from, and when the same quantity is measured for every engine, which engines are stable?
3. **RQ3 (utility).** Holding the agent, its tools, and its budget fixed, does a retrieval seed change what the agent submits, and does it change verified repair?

Contributions.

- **An exposure protocol.** A case-level ledger assigns every evaluation set one of three classes according to what development could have seen (§3). Applied to Delphi, it reclassifies the 220-case ARB partition from held-out to validation evidence; confirmatory results rest on 178 cases scored once and never before: 62 SWE-bench Verified localization instances, 18 repository-disjoint commit-to-files cases, and 98 further Verified instances drawn under a fresh salt.
- **A component-matched baseline ladder.** Four conventional retrievers add Delphi's embedding model, BM25 fusion, its cross-encoder and listwise reranker, and its query expansion one at a time, using Delphi's own prompts and parsers. The ladder attributes most of Delphi's margin over lexical baselines to the reranking head ($+0.3$ MRR) and bounds the value of everything specific to Delphi at $+0.06$–0.08 on yield metrics for issue queries (n=160) and below zero for commit-to-files queries (n=18) (§4, Figure 1).
- **Matched contracts for hosted engines.** Routing every engine's retrieved evidence—including a hosted synthesis engine's own recorded sources—through one frozen synthesis stage turns an apparent hosted lead into a four-way tie above a no-retrieval floor, and measuring retrieved-set stability for every engine shows the hosted synthesis engine to be at least as repeatable as Delphi (§5–6).
- **A preregistered executable pilot.** On the same 62 instances as the ranking evaluation, a fixed agent with a retrieval seed resolves no more verified repairs than with a random-file seed, and two trajectories per cell disagree by more than the effects under study; the pilot quantifies the sample size a decisive study needs (§7).

We claim no state of the art; the findings are largely null or negative for the engine we built, and we report them with the same prominence as the positive ones.

2 Related work

SWE-bench established executable repository-level issue resolution (Jimenez et al., 2024), while SWE-agent and Agentless expose repository navigation and localization as explicit stages (Yang et al., 2024; Xia et al., 2024). LocAgent studies a graph-guided localization agent and reports that better localization can improve multi-attempt repair (Chen et al., 2025). Repository-level code completion provides complementary evidence that cross-file retrieval matters once the edit location is known: RepoCoder alternates retrieval and generation (Zhang et al., 2023); CodeRAG adds multi-path retrieval and preference-aligned reranking (Zhang et al., 2025). Dense retrieval, late interaction, and hypothetical-document expansion motivate the components in practical context engines (Karpukhin et al., 2020; Khattab & Zaharia, 2020; Gao et al., 2023), while BM25 remains a strong exact-term baseline for code (Robertson & Zaragoza, 2009).

The closest work measures context acquisition directly. ContextBench provides human-verified file, block, and line contexts for issue-resolution tasks and scores agent trajectories (Li et al., 2026). CORE-Bench scales requirement-driven retrieval across code understanding, edit localization, and broader context (Zhang et al., 2026a). SWE-Explore derives audited line-level targets from successful trajectories (Zhang et al., 2026b). Agent Retrieval Bench (ARB) contributes workflow-derived next-context targets, base-commit hygiene, budget-aware file metrics, open-embedding and RepoMap baselines, and a closed-tool seed intervention with no-seed, random, retrieval-derived, and oracle arms (Qin & Xie, 2026). Our closed-tool experiment (§7) is a replication of that intervention with an additional seed, not a new design. What we add to this line is narrower: an exposure ledger that classifies every case by what could have influenced it, a component-matched baseline ladder, matched output contracts for hosted engines, like-for-like determinism, and a preregistered executable pilot on the same tasks as the ranking evaluation.

3 Evaluation protocol

Exposure classes. Every evaluation set is assigned a class in a case-level ledger (Appendix C, Figure 6). **C0**: never scored by any system before its single confirmatory run, drawn from a pool no development decision could see. **C1**: development. **C2**: re-partitioned from a pool whose cases were all scored in an earlier round, where that round's aggregates influenced the shipped system. The 220-case ARB partition is C2: all 427 ARB v2 cases were scored in a first development round; that round's held-out aggregates were the acceptance test for query expansion, listwise reranking, the cross-encoder blend, reciprocal-rank fusion, and the path-affinity branch (Appendix E), and its per-workflow rates motivated the quoted-path demotion fix. A second round then drew a new $25/75$ split from the same pool, excluded 50 legacy-exposed IDs, and queried the 220-case partition once. Re-drawing does not un-expose a case. Per-case outputs of the first round are not retained, so per-case tuning can neither be ruled in nor out from the archive; we therefore treat the partition as validation evidence and make no confirmatory claim on it. The C0 sets are: 62 stratified SWE-bench Verified localization instances (12 repositories; patch-marker leakage audit $0/62$); an 18-case commit-to-files final over a repository-disjoint corpus locked before any scoring; and 100 further Verified instances drawn under a fresh salt from the 438 instances no round had used (98 after two engine-agnostic rules, §4). A 40-case DS-1000 development subset drives the documentation track and is C1. The ledger controls author exposure; it cannot control whether the foundation models inside any engine saw these public repositories during training, a caveat shared by every arm.

Queries, corpus, metrics. Queries are the official ARB gold-free structured objects, serialized with sorted keys. File text comes from bare git clones at each case's base commit (materialized snapshots; binary and non-UTF-8 blobs skipped). Metrics are ARB's official per-case metrics (MRR, Recall@k) plus BCY@8k, the budget-constrained yield from packing ranked files into an 8,000-token context with ARB's official packing routine. Documentation cases score *identifier hit*: whether the returned text contains the case's gold API identifier.

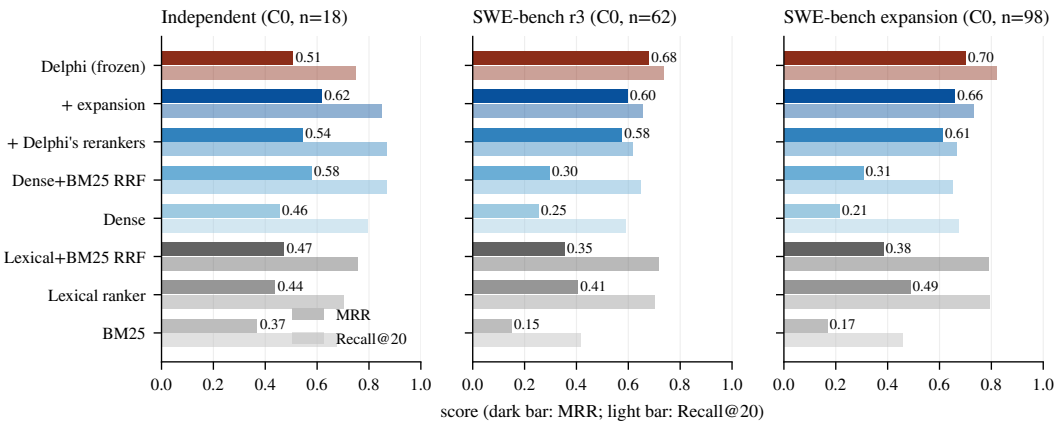

Figure 2: The three C0 sets, every system, scored once. Dark bars are MRR, light bars Recall@20. Grey: lexical comparators; blue: the matched ladder, darker as components are added; red: the frozen Delphi stack. Full tables with Recall@5, BCY@8k, any-gold counts, latencies, and paired intervals are in Appendix D (Tables 5–7).

Freeze and attestation. All Delphi numbers come from one frozen build with exact vector scan, API top-k=20, and response-level configuration attestation: every response carries the retrieval configuration actually served, and a run aborts on the first mismatch with its declared configuration. The frozen configuration is reproduced in Appendix A. Paired deltas carry cluster bootstrap 95% intervals (20,000 resamples, seed 1042; repository clusters for repository tracks, case clusters for documentation), plus exact McNemar tests on any-gold@20 movement.

Matched baseline ladder. The lexical comparators (whole-file BM25, ARB's official lexical ranker, and their development-tuned fusion) cannot say how much of a margin is Delphi's engineering and how much is a learned reranker on top of any candidate pool. We therefore add four conventional systems, each adding one Delphi component to the previous one, over the official ARB chunks (file chunk plus symbol chunks) of the identical corpus: **dense**, `text-embedding-3-small`—the embedding model inside Delphi—max-pooled to files; **hybrid**, reciprocal-rank fusion (k=60, equal weights, untuned) of the dense ranking and BM25; **+ rerankers**, the hybrid's top 50 narrowed to 30 through Delphi's cross-encoder (`ms-marco-MiniLM-L-6-v2`, blend 0.4) and then Delphi's listwise `gpt-4o` reranker over the top 20 (seed 1042, identical prompt); **+ expansion**, the same with Delphi's hypothetical-document expansion (`gpt-4o-mini`, identical prompt and gate) feeding the dense query. We call the last rung the *full conventional stack*. Prompts and parsers are imported from Delphi's source, so they are identical by construction (Appendix B). What remains between the full conventional stack and Delphi is exactly Delphi's own contribution: exact-symbol, exact-path, path-affinity, and trigram branches; its chunker and index; tuned fusion weights; quoted-path demotion. Reported latencies exclude index construction for every system.

Hosted engines. Two commercial engines are compared on the documentation track: a *documentation engine* returning raw documentation context for a library query, and a *synthesis engine* returning a grounded answer with cited sources; §6 matches their output contracts before comparing. The synthesis engine's repository surface could not be indexed at the required commits (Appendix I), so no repository number is reported for it in either direction.

Claim rule. "State of the art" would require leading every directly comparable engine on a C0 set with the complete comparable engine set runnable. No result in this paper meets it.

4 Retrieval quality under matched baselines (RQ1)

Against lexical baselines. Delphi leads whole-file BM25, the official lexical ranker, and their tuned fusion on every set (Figure 2). On the independent final the Recall@20 leads over BM25

Table 1: Pooled SWE-bench Verified localization, 160 C0 instances (Tables 6 and 7), 12 repositories. Paired Delphi−system deltas with repository-cluster bootstrap 95% intervals; * excludes zero. The last ladder rung (+ expansion) is the full conventional stack.

Delphi − system	MRR	R@5	R@20	BCY@8k	any-gold (p)
Dense (same embedding)	$+0.463^{*}$ [+0.418, +0.547]	$+0.365^{*}$ [+0.296, +0.505]	$+0.147^{*}$ [+0.086, +0.277]	$+0.490^{*}$ [+0.434, +0.568]	131 vs 108 (p=0.00043)
Dense+BM25 RRF	$+0.390^{*}$ [+0.316, +0.492]	$+0.242^{*}$ [+0.154, +0.470]	$+0.138^{*}$ [+0.090, +0.226]	$+0.440^{*}$ [+0.394, +0.546]	131 vs 110 (p=0.0011)
+ Delphi's rerankers	$+0.095^{*}$ [+0.023, +0.181]	$+0.116^{*}$ [+0.064, +0.242]	$+0.141^{*}$ [+0.084, +0.248]	$+0.108^{*}$ [+0.064, +0.188]	131 vs 110 (p=0.0011)
+ expansion	+0.057 [−0.036, +0.114]	$+0.069^{*}$ [+0.015, +0.160]	$+0.084^{*}$ [+0.013, +0.154]	$+0.076^{*}$ [+0.017, +0.128]	131 vs 120 (p=0.08)
Lexical+BM25 RRF	$+0.320^{*}$ [+0.255, +0.434]	$+0.216^{*}$ [+0.151, +0.393]	+0.026 [−0.031, +0.128]	$+0.334^{*}$ [+0.286, +0.453]	131 vs 128 (p=0.69)
Lexical ranker	$+0.237^{*}$ [+0.174, +0.391]	$+0.163^{*}$ [+0.081, +0.368]	+0.030 [−0.019, +0.117]	$+0.245^{*}$ [+0.180, +0.406]	131 vs 128 (p=0.66)
BM25	$+0.532^{*}$ [+0.491, +0.630]	$+0.494^{*}$ [+0.411, +0.738]	$+0.346^{*}$ [+0.263, +0.538]	$+0.521^{*}$ [+0.478, +0.645]	131 vs 77 (p=2.7e − 12)

(+0.074 [+0.019, +0.139]) and the lexical ranker (+0.046 [+0.009, +0.083]) exclude zero while the fusion comparison is a tie; on SWE-bench MRR leads the strongest lexical system by +0.274 [+0.194, +0.412] with Recall@20 tied; on the C2 ARB partition every lexical comparison excludes zero. None of this isolates Delphi's contribution.

Against the matched ladder. The independent final (Table 5) is the first of two clean tests. The dense rung alone, using nothing but the embedding model Delphi already calls, ties Delphi on any-gold and exceeds it on Recall@20 (0.796 vs. 0.750). Adding BM25 by untuned reciprocal-rank fusion yields MRR 0.579 and Recall@20 0.870 against Delphi's 0.508 and 0.750; the Recall@20 gap (−0.120 [−0.231, −0.019]) excludes zero. Attaching Delphi's own rerankers and expansion reaches MRR 0.619 and BCY 0.546, and Delphi's MRR deficit against the full conventional stack (−0.111 [−0.200, −0.034]) also excludes zero. Six repository clusters is a small basis for an interval and we report the direction with that caveat; the point is not that the conventional stack wins but that nothing Delphi adds on top of it is visible here.

The 62-instance SWE-bench set (Table 6) is the second clean test. Its point estimates lean the other way and its conclusion is the same. On issue text the dense rung alone is weak (MRR 0.254) and untuned fusion with BM25 helps little (0.296); Delphi's leads over both have intervals excluding zero. Attaching Delphi's rerankers to that hybrid moves MRR from 0.296 to 0.575, and expansion to 0.597: the reranking head, not the candidate branches, is where the margin over lexical systems comes from. Against the full conventional stack Delphi leads by +0.083 MRR [−0.107, +0.203], +0.080 Recall@20 [−0.105, +0.217], and +0.065 BCY [−0.062, +0.171], with any-gold 48 vs. 44 (p=0.42): a tie. On the C2 ARB partition (Table 8) Delphi's margin over the full conventional stack is +0.103 MRR [−0.008, +0.189] and +0.076 BCY [−0.019, +0.157], with Recall@5 and Recall@20 tied and any-gold 163 vs. 148 (p=0.08); against every earlier rung the leads exclude zero. That is the set on which Delphi's components were selected, so a margin there is the expected direction of selection, not evidence. Its workflow breakdown is informative as a hypothesis (Appendix D, Table 10): Delphi's lead is concentrated in `trace2code` (MRR 0.579 vs. 0.149) and `edit2ripple`, whose queries carry file paths and symbol names that its exact-path, path-affinity, and symbol branches match directly, while on `code2test` and `comment2context` the full conventional stack is equal or ahead. An exploratory split of the 62 SWE-bench issues by whether they contain a traceback or a Python path does not reproduce that pattern (Delphi's largest advantage is on issues with neither, n=35), so we record it as a hypothesis for a designed test.

The expansion, and the pooled picture. To widen the C0 basis we drew 100 further Verified instances under a fresh salt from the 438 no round had used, applied two engine-agnostic rules before any scoring (a gold file must exist at the base commit, which removed one patch-created file from one instance's gold set; ARB's leakage check, which excluded two instances whose issue text carried patch markers), provisioned the frozen build on the 98 remaining snapshots (index audit: 182,415 files, 1,079,743 chunks and embeddings, zero mismatches), and scored every system once (Table 7). The shape repeats: dense and untuned hybrid are weak on issue text, the reranking head does most of the work (0.306 → 0.611 MRR), expansion adds a little (0.660), and Delphi finishes at 0.700. On this larger set Delphi's margin over the full conventional stack is +0.041 MRR [−0.028, +0.112], +0.061 Recall@5 [+0.023, +0.148], +0.087 Recall@20 [+0.042, +0.167], and +0.082 BCY [+0.030, +0.152]: three of four intervals exclude zero. Pooling the 160 SWE-bench instances (Table 1, Figure 3) gives the same reading, +0.057 MRR [−0.036, +0.114] with Recall@5, Recall@20, and BCY all excluding zero and any-gold 131 vs. 120 (p=0.08). Against the lexical ranker, which is strong at Recall@20 on issue text (0.793), Delphi's pooled Recall@20 lead

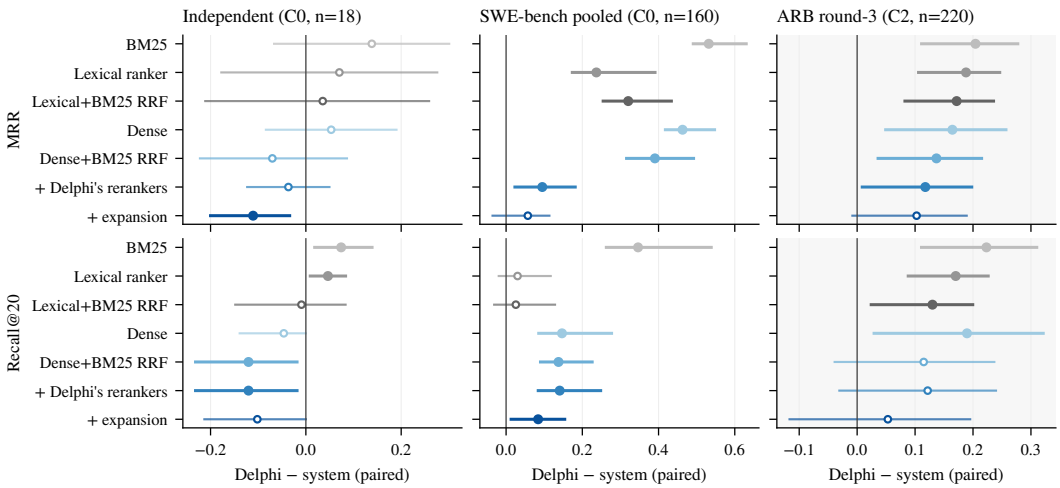

Figure 3: Paired Delphi−system differences with repository-cluster bootstrap 95% intervals (filled markers: interval excludes zero). Left: the 18-case independent final. Centre: the 160 pooled SWE-bench instances. Right, shaded: the C2 ARB partition, validation only. Recall@5 and BCY@8k panels are in Figure 7.

is a tie ($+0.030$ $[-0.019, +0.117]$) while its MRR and BCY leads exclude zero. The rank distribution (Figure 8) shows where the difference sits: Delphi and the full conventional stack place a gold file first on 62% and 58% of pooled instances, and Delphi misses the top 20 on 18% against 25%; the lexical ranker misses on 20% but ranks gold first on only 34%.

The answer to RQ1 is therefore not one number. On SWE-bench-style issue queries Delphi's own additions are worth roughly $+0.06$ to $+0.09$ on the yield metrics over a conventional stack built from its own parts, resolved at n=160; on the commit-to-files set they are worth less than nothing, resolved at n=18. In both cases the component that matters most is the one any practitioner would add first: a reranking head over a dense-plus-lexical candidate pool, which moves MRR by $+0.3$ where Delphi's remaining engineering moves it by $+0.06$. Latency is the price: 2–6 s per query for Delphi and the two LLM rungs against 0.5 s for dense retrieval (Table 11). Development-time ablations and bug fixes, all selected on development or C2 evidence, are reported in Appendix E.

5 Determinism, like for like (RQ2)

The pipeline was the variance. Repeating eight ARB queries ten times against the pre-freeze stack gave a mean exact top-10 rate of 30.6%. Paired traces isolated uncached, unseeded LLM stages; insertion-order ties in fusion SQL; and approximate nearest-neighbour search under concurrency. Total SQL ordering, single-flight sharing of concurrent identical hosted calls, an explicit seed, and a persistent stage cache take exact repeats to 100% in-process across two concurrent 8×10 packets and two 75-case repeats; a genuine cold restart still moved $7/8$ lists until the cache was warm, after which $8/8$ survive two restarts. We claim restart-durable repeatability from the first persisted answer, not that two clean installations agree on their first answer. Embedding APIs were not the problem: the small OpenAI model jitters at the fifth decimal and never changed top-ten membership across N=20 repeats; Gemini embeddings were bitwise deterministic (Appendix G).

Same quantity, every engine. A naive comparison—98.0% byte-identical context for Delphi against 35.6% and 0.0% for the hosted engines—compares retrieval stability for two engines with generated-text stability for the third. Table 2 separates the layers. On retrieved-set stability the synthesis engine's cited sources are identical across every pair (1.000), Delphi's retrieved items across 98.0%, and the documentation engine's across 54% (quality-guided) or 50% (oracle IDs). What differs for the synthesis engine is the layer above retrieval: its answer text never repeats byte-for-byte, and that variation flips the identifier-hit outcome in 7.1% of pairs, against 0% for Delphi. The defensible statements are narrow: Delphi's caching and total ordering make its retrieved text repeatable,

Table 2: Like-for-like repeatability on ten documentation cases × ten runs (450 run pairs per engine). Retrieved-set columns use each engine's observable retrieval unit: retrieved items for Delphi and the documentation engine, the set of cited URLs for the synthesis engine. Byte-exact is the delivered context text; hit agreement is whether the identifier-hit outcome agreed across the pair.

Engine	Set exact	Order exact	Set Jaccard	Byte exact	Hit agreement
Delphi (raw context, k=5)	0.980	0.980	0.993	0.980	1.000
Documentation engine (quality-guided IDs)	0.542	0.362	0.835	0.356	0.918
Documentation engine (oracle IDs)	0.496	0.258	0.806	0.253	0.909
Synthesis engine (cited-URL set)	1.000	1.000	1.000	0.000	0.929

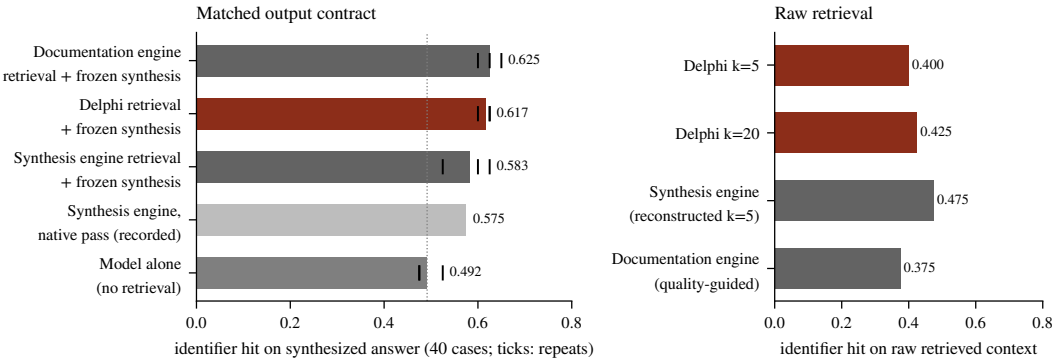

Figure 4: Documentation track, 40 development cases (C1). Left: every engine's retrieved evidence—including the synthesis engine's own five recorded sources—routed through one frozen synthesis stage (`gpt-5.4-mini`, one prompt, 8,000-token budget), three repeats (ticks); the dotted line is the no-retrieval floor. Right: the raw retrieved context scored directly. Tables with intervals are in Appendix F.

and repeatable retrieved text makes downstream decisions repeatable; the documentation engine's retrieval drifts; the synthesis engine's retrieval does not drift, its prose does. A caching wrapper would likely give the hosted engines the same repeatability; we did not test it.

6 Documentation retrieval under matched output contracts

Documentation retrieval is where hosted engines are strongest and where output contracts differ most. The synthesis engine returns a fluent answer with citations; the documentation engine and Delphi return raw context. Scored naively, the synthesis engine leads (identifier hit 0.575 against Delphi's 0.400 and the documentation engine's 0.375). That comparison conflates retrieval with the synthesis model's parametric knowledge.

Matching the contract. Routing only the two raw-context engines through a frozen answer-style synthesis stage (`gpt-5.4-mini`, 1,600 output tokens, one system prompt) and comparing them with the synthesis engine's *native* pass aligns the output format but not the synthesis model or prompt, and is only partially matched. The synthesis engine's raw-retrieval mode returned no sources during the recorded round, so its retrieval could not be routed directly; its synthesized responses, however, carry the five retrieved source chunks it grounded on. We reconstructed that context from the recorded responses (five sources per case, mean 2,754 tokens), packed it under the same 8,000-token budget with the same routine as every other arm, and routed it through the same frozen stage, three repeats, no new hosted calls. Figure 4 is a four-arm comparison under one synthesis model, one prompt, one budget.

Three findings follow. First, on raw retrieved context the synthesis engine is ahead of Delphi: its reconstructed five-source context hits the gold identifier on 0.475 of cases against Delphi's 0.400 (k=5) and 0.425 (k=20). Second, under the matched contract the four arms finish within 0.05 of one another and every engine-versus-engine interval includes zero (Table 13); the synthesis engine's

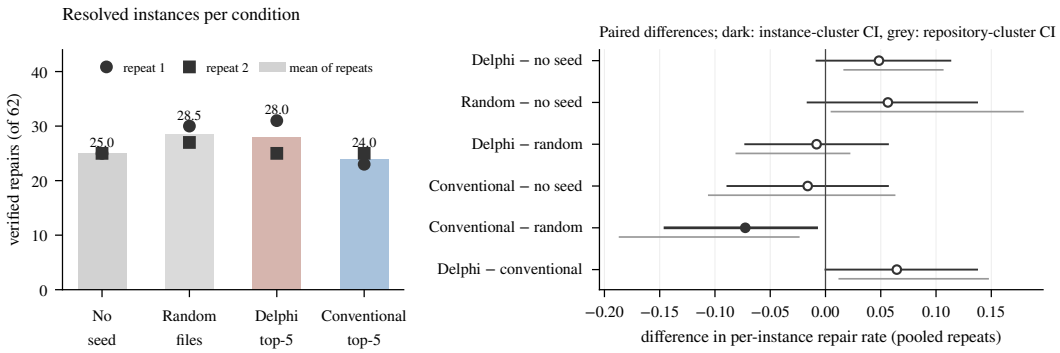

Figure 5: Executable pilot: 62 C0 SWE-bench Verified instances, mini-SWE-agent with `gpt-5.4-mini`, step budget 50, two independent trajectories per instance and condition. Left: verified repairs per condition and repeat. Right: paired differences in per-instance repair rate with instance-cluster (dark) and repository-cluster (grey) bootstrap intervals; the filled marker is the only interval excluding zero. Per-repeat tables and the protocol are in Appendix H.

matched arm (0.583) is within 0.008 of its native pass, so its own synthesis added nothing the frozen stage did not. Third, the metric saturates: the bare model scores 0.492, so roughly 80% of every arm's score is parametric knowledge, and the retrieval-over-floor deltas are +0.13, +0.13, and +0.09. We detect no statistically resolved difference between any two engines under this evaluation; "parity or better" is not supported either, since an interval of [−0.09, +0.16] does not establish it, and non-inferiority would require a predeclared margin and a sample several times this size.

7 Agent utility (RQ3)

Closed-tool seeding (a replication). ARB's seed intervention holds a closed-tool agent fixed and varies its initial context. We replicated it with the policy model `gpt-5.4-mini`, tools `list_dir`/`grep`/`read_file`/`submit`, and 40 paired development cases per arm, adding a seed from the pre-freeze Delphi. File-F1: no seed 0.046; random files 0.013; BM25 seed 0.093; lexical seed 0.119; Delphi seed 0.234; oracle 0.858. The Delphi seed beats no-seed by +0.188 [0.055, 0.294] and BM25 by +0.142 [0.059, 0.208]; against lexical the F1 interval includes zero [−0.002, 0.190]. Seeds change which files a frozen agent submits, ordered by seed quality, replicating ARB's finding. The oracle ceiling shows localization headroom in this harness; it does not show that retrieval dominates reasoning, editing, or validation failures in a repairing agent.

DS-1000 (unresolved, not negative). With the frozen generator, 40 cases × 3 repeats: no retrieval 0.575; documentation engine 0.567 (−0.008 [−0.142, 0.125]); Delphi retrieval+synthesis 0.592 (+0.017 [−0.125, 0.158]); synthesis engine 0.642 (+0.067 [−0.075, 0.200]). Every interval spans effects from substantial harm to substantial benefit. This experiment does not show that retrieval fails to help; it shows that 40 cases cannot tell.

Executable pilot (preregistered). To measure retrieval and executable outcome on the same tasks, we preregistered a pilot (Appendix H) on the 62 C0 SWE-bench instances whose rankings every system in Table 6 had already produced once. The agent is mini-SWE-agent with `gpt-5.4-mini`, ordinary shell tools in every condition, and a fixed step budget; a seed is a block appended to the issue naming the top five files a recorded ranking produced, with the head of each file, capped at 3,000 tokens. Conditions: no seed; five random non-gold files in the same block (prompting control); Delphi's recorded top five; the full conventional stack's recorded top five. The outcome is verified repair under the official harness, paired by instance, with instance- and repository-cluster intervals. Every condition was run twice (two independent trajectories per instance); Figure 5 reports both repeats and their pooled per-instance means.

The two repeats disagree by more than the effects under study, which is the first result. Resolved counts were 25/30/31/23 (none/random/Delphi/conventional) in the first repeat and 25/27/25/25 in

the second: the Delphi seed's six-instance lead over no seed in the first repeat vanished in the second, and the conventional seed's eight-instance deficit shrank to zero. Pooled, no seed effect is resolved. A Delphi seed changes the per-instance repair rate by $+0.048$ $[-0.008, +0.113]$ against no seed, and by -0.008 $[-0.073, +0.056]$ against a block of five random non-gold files, which itself sits at $+0.056$ $[-0.016, +0.137]$ over no seed. Whatever the block contributes at this size, it is not separable from the value of naming the right files. The conventional stack's seed, whose top five named a gold file on 42 instances against Delphi's 47, is the only arm that trails a comparator with an interval excluding zero (-0.073 $[-0.145, -0.008]$ against the random control); we have no explanation and offer none. Budgets were not binding: agents used 7–8 of 50 steps and cost about two cents per instance; the seed's tokens raised cost by roughly 10–30%, and Delphi's seed reduced steps slightly relative to the random block. The pilot's width is its finding: at n=62 with two trajectories, effects of ± 0.05 in verified repair are inside sampling noise, so a decisive study needs several hundred instances and more repeats, which the per-run cost (about $11 for all 496 trajectories) makes affordable.

8 Discussion and limitations

Supported. (1) Delphi leads whole-file BM25 and the official lexical ranker on every set, with intervals excluding zero on the C0 sets for Recall@20 (independent) and for MRR and BCY (SWE-bench, 160 pooled); its Recall@20 lead over the lexical ranker on SWE-bench is a tie. (2) Against a conventional dense+BM25 hybrid built from Delphi's own embedding model with Delphi's own rerankers and expansion attached, Delphi leads by $+0.06$–0.08 on Recall@5, Recall@20, and BCY over 160 pooled SWE-bench instances (intervals excluding zero; MRR tied), and trails on the 18-case independent final (two intervals excluding zero). The reranking head accounts for most of the margin over lexical systems on every set. (3) Under one frozen synthesis stage, no two documentation engines are statistically distinguishable at n=40, and all sit 0.08–0.13 above a no-retrieval floor. (4) Delphi's retrieved text is repeatable in-process and across restarts with a warm cache; on retrieved-set stability the hosted synthesis engine is at least as stable. (5) Retrieval seeds change a closed-tool agent's file selection, ordered by seed quality, replicating ARB.

Not supported. (1) State of the art, anywhere. (2) Any confirmatory result on the ARB partition, which is C2. (3) Documentation parity or non-inferiority. (4) Downstream utility on DS-1000 at n=40, or in executable repair at n=62, in either direction. (5) That Delphi's structural branches, tuned weights, or quoted-path demotion add value in general: they separate from the full conventional stack on issue-style queries and not on commit-to-files queries, and the mechanism is untested.

Limitations. One engine is studied in depth; the hosted engines are studied only where their surfaces permitted a matched measurement, and one arm is missing. The C0 sets are small (18, 62, 98) and drawn from two query styles; the independent final has six repository clusters. Every arm shares one embedding model and one family of foundation models, so training-set exposure of the public repositories is uncontrolled. The executable pilot uses a small policy model and a single agent scaffold, and its repeats show that two trajectories per cell are too few. The documentation metric is a surface identifier match, which saturates. The exposure ledger records what the authors could see; it does not audit what the models had seen.

9 Conclusion

Measured against baselines built from its own parts, and on cases its development could not see, most of Delphi's advantage is the part any careful practitioner would assemble: an embedding model, BM25, reciprocal-rank fusion, a small cross-encoder, a listwise reranker, and query expansion. What is Delphi's own is worth a further $+0.06$ to $+0.08$ on yield metrics for issue-style queries at n=160, and nothing on commit-to-files queries at n=18. Margins measured on the re-partitioned ARB set do not survive the ledger that classifies how that set was used, and margins over hosted engines do not survive matching the quantity being measured. What survives is the protocol—case-level exposure classes, component-matched baselines, matched output contracts, like-for-like determinism—and the open question the pilot begins to answer: whether any of this changes what an agent repairs.

LLM USAGE STATEMENT

Generative AI tools were used to help formulate experimental questions, write and test evaluation and product code, operate experiments, inspect failures, search for related work, and draft and edit this manuscript. They were not used to generate benchmark gold labels. Empirical claims are checked against persisted run artifacts and canonical repository snapshots; every table and figure in this paper is generated from those artifacts by scripts in the released archive. All AI-assisted work remains subject to human review before submission, and the authors take responsibility for the final content, claims, code, and artifacts.

ETHICS STATEMENT

The study uses public open-source repositories and public benchmark records. Repository contents are processed locally except where a named hosted retrieval or model API is itself the evaluated system. We retain commit identifiers and public issue or review text for provenance, but do not intentionally collect private repository data or credentials. Because benchmark results can be used in product marketing, we predeclare the claim rule, preserve failed runs, publish the exposure ledger, and give losses the same prominence as wins.

REPRODUCIBILITY STATEMENT

The archive accompanying this paper (Appendix K) contains the split manifests and case-level exposure ledger (Appendix C), every per-case artifact (queries, gold paths, ranked outputs, synthesized answers, latencies), every paired-analysis result, the implementation of every system and comparator including the matched ladder (Appendix B), the frozen serving configuration (Appendix A), the pilot protocol, datasets, and all 496 agent trajectories with harness verdicts (Appendix H), the prompts used by every LLM stage verbatim (Appendices B, F, H), and the scripts that regenerate every table and figure in this paper from those artifacts. Repository corpora are re-cloned from public GitHub at the recorded commits by a provided script; hosted-engine runs require the reader's own credentials.

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APPENDIX

The appendix documents every experiment in enough detail to rerun it from the released archive. Appendix A gives the frozen Delphi configuration; B the matched baseline ladder and the verbatim prompts it shares with Delphi; C the exposure ledger and the round chronology; D the complete repository-retrieval tables and additional figures; E the development-time ablations and the rejected ideas; F the documentation track; G the determinism probes; H the executable pilot protocol, seed block, and per-repeat results; I the hosted engine accounting; J compute and cost; and K the artifact release.

A FROZEN SYSTEM CONFIGURATION

All Delphi numbers come from a single frozen build (its revision identifier is recorded in the archive), served natively against PostgreSQL 14 with pgvector. Every scored response carried the following attested configuration (Table 3): embedding model `text-embedding-3-small`; vector mode exact scan (approximate search disabled for scoring); hybrid candidate branches vector, BM25, exact symbol, exact path, path affinity, and trigram with reciprocal-rank fusion weights $0.50/0.25/0.20/0.15/0.15/0.05$ and k=60; 50 fused candidates; cross-encoder `cross-encoder/ms-marco-MiniLM-L-6-v2` over the top 30 with blend α=0.4 (code-aware reranker disabled); quoted-path demotion when the query envelope carries evidence keys; listwise reranking of the top 20 by `gpt-4o` at temperature 0, seed 1042, 280-character excerpts, 300-token replies; hypothetical-document expansion by `gpt-4o-mini` (temperature 0, seed 1042, 220-token replies) gated on prose-like queries and feeding only the vector branch; persistent stage cache; API top-k=20; file-diverse BM25 and path-token branches disabled; agent quality mode with tests, docs, examples, configs, and manifests indexed and generated source retained under 500,000-byte and NUL-content guards. Context packing for BCY@8k uses ARB's official packing routine at an 8,000-token budget.

Table 3: Frozen serving configuration of Delphi. Secrets and local paths omitted.

Setting	Value
Embedding provider and model	OpenAI, `text-embedding-3-small` (1,536 dimensions)
Vector search	exact scan (HNSW index present but not consulted)
Candidate branches	vector, BM25, exact symbol, exact path, path affinity, trigram
Fusion	reciprocal rank, k=60, weights $0.50/0.25/0.20/0.15/0.15/0.05$
Fused candidates / rerank window	50 / 30
Cross-encoder	`cross-encoder/ms-marco-MiniLM-L-6-v2`, blend α=0.4
Code-aware reranker	disabled
Query expansion	enabled; `gpt-4o-mini`, temperature 0, seed 1042, gated on prose-like queries
Listwise reranking	enabled; `gpt-4o`, top 20, temperature 0, seed 1042
LLM stage cache	persistent (archived) with a 2,048-entry in-memory tier
Quoted-path demotion	enabled when the query envelope carries evidence keys
File-diverse BM25 / path-token branches	disabled
API top-k	20
Worker threads	4
Rate limits	disabled for local scoring

Attestation. Every search response from the frozen build carries a configuration object listing the branches, weights, reranker, listwise, expansion, vector mode, and k actually used to serve it. The evaluation runner compares this object with the run's declared configuration on every response and aborts the run on the first mismatch; each run summary records the verification outcome and the set of observed configurations, which is a singleton in every reported run.

Index audit. Before the expansion set was scored, an audit walked every provisioned snapshot and compared the file count at the base commit (after the same size and binary guards) with the indexed file, chunk, and embedding counts in the database: 98 snapshots, 182,415 files, 1,079,743 chunks, 1,079,743 embeddings, zero mismatches. A gold-searchability preflight then confirmed that every gold path of every case is present in the index; this preflight surfaced the one patch-created gold file (§4).

B MATCHED BASELINE LADDER

Corpus and chunks. Every rung operates on the same materialized snapshot as Delphi and the lexical comparators, chunked with the official ARB routine (one whole-file chunk plus one chunk per top-level symbol). Non-UTF-8 and binary blobs are skipped by the same guard Delphi uses.

Dense. Chunks are embedded with `text-embedding-3-small` (1,536 dimensions) in token-aware batches (at most 200,000 tokens or 512 texts per request; texts above 8,000 tokens are truncated with the `cl100k_base` tokenizer); vectors are cached keyed by model and content hash, so a chunk is embedded once regardless of how many rungs read it. Queries are embedded with the same model; similarity is cosine; a file's score is the maximum over its chunks; the top 50 files are returned.

Hybrid. ARB's official BM25 chunk ranker over the same chunks, max-pooled to files, fused with the dense file ranking by reciprocal-rank fusion, $\mathrm{RRF}(f) = \sum_r 1/(60 + \mathrm{rank}_r(f))$ with equal weights. Nothing is tuned.

+ Delphi's rerankers. The top 50 fused files are narrowed to 30 with Delphi's cross-encoder (`ms-marco-MiniLM-L-6-v2`, scored on the query and the first 4,000 characters of the file's best-matching chunk, its sigmoid blended at α=0.4 with the fused score normalized by the top fused score, as in Delphi), then the top 20 are reordered by Delphi's listwise reranker (`gpt-4o`, temperature 0, seed 1042, 280-character excerpts). The system prompt and the reply parser are imported from Delphi's listwise-reranking module at the frozen revision.

+ Delphi's expansion. Delphi's hypothetical-document expansion (`gpt-4o-mini`, temperature 0, seed 1042, 220-token reply) is applied under Delphi's own gate (prose-like queries only) and the expansion text is appended to the dense query; BM25 sees the original query, as in Delphi. The prompt is imported from Delphi's query-expansion module.

Caching and latency. LLM replies are cached keyed by model, prompt, and seed, so a repeat of a rung is bitwise identical and free. Reported latencies are wall-clock per query on a warm process excluding index construction, as for Delphi, and include hosted API round trips.

Listwise reranker system prompt (verbatim, shared with Delphi).

```
You rank candidate source files by how directly each one
answers a developer's request about a codebase.  You are
given a request and a numbered list of candidates, each with
its repository path and an excerpt.  Return the candidate
numbers ordered best first, as a JSON array of integers, and
nothing else.  Example:  [4,1,9,2] Include every candidate
number exactly once.  Judge by what the request actually
asks for.  If it asks which test covers some behaviour, a
test file is the answer and the implementation is not.  If
it asks where something is implemented, the reverse holds.
Prefer the file that would have to be opened or edited to
satisfy the request over one that merely mentions the same
words.
```

Query-expansion system prompt (verbatim, shared with Delphi).

```
You write short, plausible code snippets that would answer a
developer's question about an unfamiliar codebase.  Reply
with code only:  no prose, no explanation, no fences.
Invent idiomatic function, class, and variable names for
the language implied by the question.  Being wrong about the
specifics is fine; using the vocabulary real code would use
is what matters.  Keep it under 15 lines.
```

C EXPOSURE LEDGER AND CHRONOLOGY

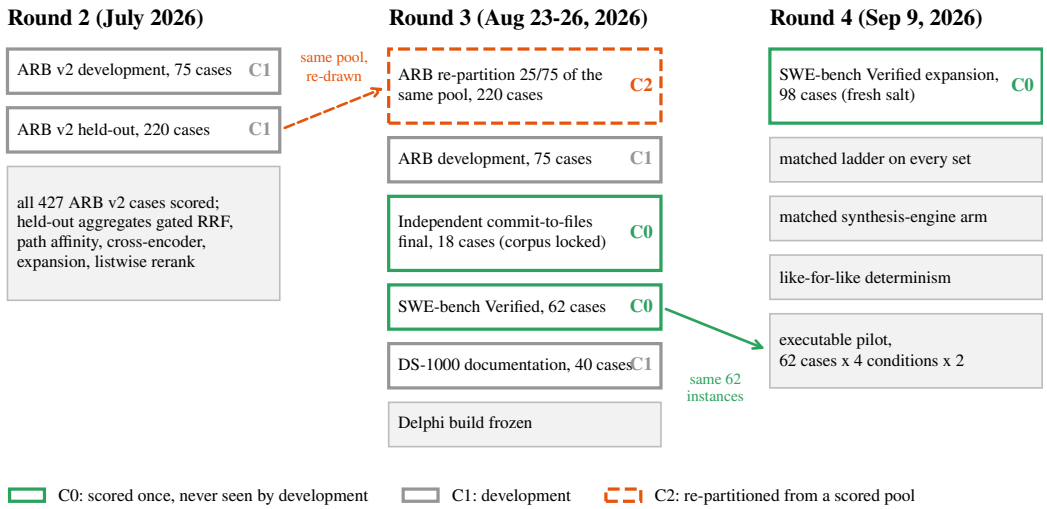

Figure 6: Evaluation rounds, the sets each round scored, and their exposure class. Round 2 scored every case of the ARB v2 pool and used the held-out aggregates to accept components; round 3 re-drew a 220-case partition from that pool (C2) and introduced the C0 sets; round 4 added the matched ladder, the 98-case expansion, the matched synthesis-engine arm, like-for-like determinism, and the executable pilot on the round-3 62-instance set.

Table 4: Case-level exposure ledger. Every case ID of every set is listed in the machine-readable ledger in the archive with the evidence for its class.

Set	Cases	Class	Supports
ARB round-3 partition (positive workflows)	220	C2	development/validation evidence; not a confirmatory test of the frozen stack
ARB round-3 development	75	C1	development
Independent commit-to-files final	18	C0	confirmatory
Independent commit-to-files development	48	C1	development
SWE-bench Verified, round 3	62	C0	confirmatory
SWE-bench Verified, round-4 expansion	98	C0	confirmatory
DS-1000 documentation subset	40	C1	development

Decisions and the evidence that informed them. Query-time embedding alignment, RRF fusion, the path-affinity branch, the cross-encoder blend, expansion, and listwise reranking were accepted on round-2 development (75) and round-2 held-out (220) aggregates over the ARB v2 pool that round 3 re-partitioned (Appendix E); quoted-path demotion on round-2 per-workflow rates over all 427 cases; exact scan, single-flight, the persistent cache, and total SQL ordering on round-3 development and determinism probes; generated-source indexing on the gold-path preflight over the independent and ARB development sets; the File-Okapi rejection on both development sets. No decision was informed by any C0 set: the independent final corpus was locked before any query was issued; the 62 SWE-bench instances and the 98 expansion instances were each scored once per system after the build was frozen.

Why re-drawing does not un-expose. A case scored in round 2 contributed to the aggregate that accepted or rejected a component. Any partition drawn later from the same pool inherits that influence at the aggregate level, whatever the seed. What a fresh draw does exclude is per-case tuning, and only if per-case outputs were never inspected; the round-2 per-case outputs are not in the archive, so this cannot be demonstrated from the record, and we do not claim it.

What the ledger cannot control. The foundation models inside every arm (OpenAI embeddings, `gpt-4o`, `gpt-4o-mini`, `gpt-5.4-mini`, and the models behind the hosted engines) were trained on public code that very likely includes these repositories and possibly the fixing commits. This affects every arm and every rung equally within an experiment, but it means absolute numbers are not estimates of performance on unseen code.

D COMPLETE REPOSITORY-RETRIEVAL RESULTS

Table 5: Independent commit-to-files final (C0): 18 cases, 6 repositories, scored once. Any-gold is the number of cases with a gold file in the top 20; latency is the median query time in seconds. Bold marks the best value in each metric column. Δ rows are Delphi minus the named system with repository-cluster bootstrap 95% intervals; * excludes zero. The last ladder rung (+ expansion) is the full conventional stack.

System	MRR ↑	R@5 ↑	R@20 ↑	BCY@8k ↑	any-gold ↑	latency (s) ↓
Delphi (frozen)	0.508	0.551	0.750	0.454	15/18	3.7
Dense (same embedding)	0.455	0.523	0.796	0.347	15/18	0.5
Dense+BM25 RRF	0.579	0.639	**0.870**	0.403	16/18	0.2
+ Delphi's rerankers	0.545	0.644	**0.870**	0.454	16/18	2.3
+ expansion	**0.619**	**0.662**	0.852	**0.546**	16/18	4.1
Lexical+BM25 RRF	0.473	0.500	0.759	0.287	15/18	1.6
Lexical ranker	0.438	0.403	0.704	0.259	14/18	1.1
BM25	0.370	0.458	0.676	0.259	14/18	1.2
*Paired Delphi−system deltas, repository-cluster bootstrap 95% intervals; * interval excludes zero*						
Δ vs. Dense (same embedding)	+0.053 [−0.084, +0.190]	+0.028 [−0.148, +0.194]	−0.046 [−0.139, +0.000]	+0.106 [−0.074, +0.259]	15 vs 15	p=1
Δ vs. Dense+BM25 RRF	−0.070 [−0.222, +0.086]	−0.088 [−0.236, +0.056]	−0.120* [−0.231, −0.019]	+0.051 [−0.120, +0.222]	15 vs 16	p=1
Δ vs. + Delphi's rerankers	−0.037 [−0.123, +0.049]	−0.093 [−0.324, +0.083]	−0.120* [−0.231, −0.019]	+0.000 [−0.139, +0.194]	15 vs 16	p=1
Δ vs. + expansion	−0.111* [−0.200, −0.034]	−0.111 [−0.324, +0.046]	−0.102 [−0.213, +0.000]	−0.093 [−0.278, +0.074]	15 vs 16	p=1
Δ vs. Lexical+BM25 RRF	+0.035 [−0.211, +0.259]	+0.051 [−0.088, +0.194]	−0.009 [−0.148, +0.083]	+0.167 [−0.074, +0.472]	15 vs 15	p=1
Δ vs. Lexical ranker	+0.070 [−0.177, +0.276]	+0.148 [−0.046, +0.343]	+0.046* [+0.009, +0.083]	+0.194 [−0.046, +0.481]	15 vs 14	p=1
Δ vs. BM25	+0.138 [−0.067, +0.301]	+0.093 [−0.037, +0.278]	+0.074* [+0.019, +0.139]	+0.194 [−0.046, +0.481]	15 vs 14	p=1

Table 6: SWE-bench Verified localization, round 3 (C0): 62 instances, 12 repositories, scored once. Columns as in Table 5.

System	MRR ↑	R@5 ↑	R@20 ↑	BCY@8k ↑	any-gold ↑	latency (s) ↓
Delphi (frozen)	**0.680**	**0.704**	**0.737**	**0.638**	48/62	3.6
Dense (same embedding)	0.254	0.324	0.591	0.210	39/62	0.5
Dense+BM25 RRF	0.296	0.452	0.648	0.234	43/62	2.0
+ Delphi's rerankers	0.575	0.598	0.616	0.533	41/62	3.7
+ expansion	0.597	0.623	0.657	0.573	44/62	6.2
Lexical+BM25 RRF	0.354	0.500	0.719	0.306	47/62	13.2
Lexical ranker	0.406	0.509	0.702	0.387	46/62	11.0
BM25	0.149	0.230	0.416	0.145	29/62	10.7
*Paired Delphi−system deltas, repository-cluster bootstrap 95% intervals; * interval excludes zero*						
Δ vs. Dense (same embedding)	+0.426* [+0.328, +0.492]	+0.380* [+0.262, +0.583]	+0.146 [+0.000, +0.288]	+0.427* [+0.340, +0.500]	48 vs 39	p=0.064
Δ vs. Dense+BM25 RRF	+0.384* [+0.275, +0.464]	+0.252* [+0.152, +0.442]	+0.089 [−0.038, +0.182]	+0.404* [+0.288, +0.517]	48 vs 43	p=0.3
Δ vs. + Delphi's rerankers	+0.104 [−0.065, +0.210]	+0.106 [+0.000, +0.263]	+0.121 [−0.033, +0.259]	+0.105 [+0.000, +0.190]	48 vs 41	p=0.14
Δ vs. + expansion	+0.083 [−0.107, +0.203]	+0.081 [−0.056, +0.251]	+0.080 [−0.105, +0.217]	+0.065 [−0.062, +0.171]	48 vs 44	p=0.42
Δ vs. Lexical+BM25 RRF	+0.325* [+0.216, +0.453]	+0.204* [+0.099, +0.455]	+0.018 [−0.073, +0.075]	+0.331* [+0.211, +0.552]	48 vs 47	p=1
Δ vs. Lexical ranker	+0.274* [+0.194, +0.412]	+0.195* [+0.107, +0.403]	+0.035 [−0.042, +0.111]	+0.251* [+0.157, +0.433]	48 vs 46	p=0.75
Δ vs. BM25	+0.530* [+0.450, +0.662]	+0.474* [+0.352, +0.726]	+0.321* [+0.189, +0.572]	+0.493* [+0.395, +0.669]	48 vs 29	p=0.00055

Table 7: SWE-bench Verified expansion (C0): 98 further instances drawn under a fresh salt and never used in any round, 11 repositories, scored once from the frozen build after the index audit and the gold-searchability preflight. Columns as in Table 5.

System	MRR ↑	R@5 ↑	R@20 ↑	BCY@8k ↑	any-gold ↑	latency (s) ↓
Delphi (frozen)	**0.700**	**0.745**	**0.820**	**0.685**	83/98	6.4
Dense (same embedding)	0.214	0.389	0.673	0.156	69/98	0.8
Dense+BM25 RRF	0.306	0.509	0.651	0.223	67/98	2.0
+ Delphi's rerankers	0.611	0.622	0.667	0.576	69/98	3.8
+ expansion	0.660	0.684	0.733	0.603	76/98	6.2
Lexical+BM25 RRF	0.383	0.521	0.789	0.350	81/98	4.2
Lexical ranker	0.487	0.603	0.793	0.445	82/98	2.2
BM25	0.168	0.238	0.457	0.146	48/98	2.2
*Paired Delphi−system deltas, repository-cluster bootstrap 95% intervals; * interval excludes zero*						
Δ vs. Dense (same embedding)	+0.486* [+0.434, +0.622]	+0.356* [+0.281, +0.496]	+0.147* [+0.083, +0.330]	+0.529* [+0.453, +0.656]	83 vs 69	p=0.0043
Δ vs. Dense+BM25 RRF	+0.395* [+0.299, +0.552]	+0.236* [+0.133, +0.513]	+0.168* [+0.116, +0.307]	+0.463* [+0.391, +0.636]	83 vs 67	p=0.0015
Δ vs. + Delphi's rerankers	+0.089* [+0.022, +0.227]	+0.123* [+0.067, +0.277]	+0.153* [+0.099, +0.298]	+0.110* [+0.062, +0.238]	83 vs 69	p=0.0043
Δ vs. + expansion	+0.041 [−0.028, +0.112]	+0.061* [+0.023, +0.148]	+0.087* [+0.042, +0.167]	+0.082* [+0.030, +0.152]	83 vs 76	p=0.17
Δ vs. Lexical+BM25 RRF	+0.317* [+0.229, +0.466]	+0.224* [+0.141, +0.415]	+0.031 [−0.045, +0.205]	+0.335* [+0.247, +0.472]	83 vs 81	p=0.79
Δ vs. Lexical ranker	+0.213* [+0.135, +0.404]	+0.142* [+0.050, +0.396]	+0.027 [−0.035, +0.163]	+0.241* [+0.150, +0.439]	83 vs 82	p=1
Δ vs. BM25	+0.533* [+0.453, +0.668]	+0.507* [+0.405, +0.750]	+0.362* [+0.279, +0.554]	+0.539* [+0.455, +0.712]	83 vs 48	p=5.5e − 10

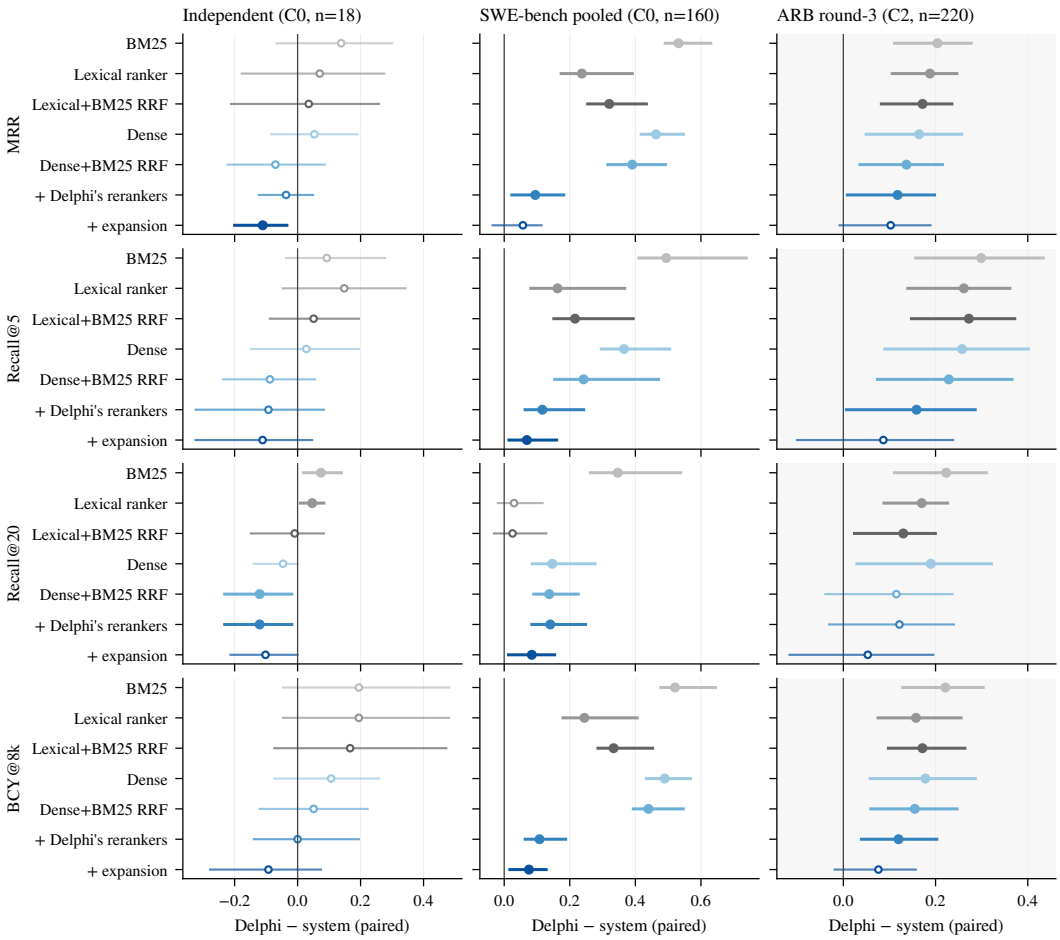

Figure 7: Paired Delphi−system differences on all four metrics with repository-cluster bootstrap 95% intervals (20,000 resamples, seed 1042). Filled markers: interval excludes zero. Left: independent final (C0, n=18, 6 clusters). Centre: pooled SWE-bench (C0, n=160, 12 clusters). Right, shaded: ARB round-3 partition (C2, n=220, 19 clusters), validation only.

Table 8: ARB round-3 partition (C2, validation evidence only): 220 cases, 19 repositories. Columns as in Table 5. Reported for completeness; no confirmatory claim rests on it.

System	MRR ↑	R@5 ↑	R@20 ↑	BCY@8k ↑	any-gold ↑	latency (s) ↓
Delphi (frozen)	**0.332**	**0.476**	**0.648**	**0.296**	163/220	4.3
Dense (same embedding)	0.168	0.218	0.458	0.118	115/220	0.5
Dense+BM25 RRF	0.195	0.247	0.533	0.141	137/220	0.5
+ Delphi's rerankers	0.214	0.317	0.526	0.176	135/220	2.1
+ expansion	0.229	0.389	0.595	0.220	148/220	4.2
Lexical+BM25 RRF	0.160	0.204	0.518	0.124	131/220	1.9
Lexical ranker	0.144	0.215	0.478	0.138	123/220	1.4
BM25	0.128	0.177	0.425	0.075	113/220	1.3
*Paired Delphi−system deltas, repository-cluster bootstrap 95% intervals; * interval excludes zero*						
Δ vs. Dense (same embedding)	$+0.164^{*}$ [+0.049, +0.257]	$+0.258^{*}$ [+0.090, +0.401]	$+0.190^{*}$ [+0.029, +0.321]	$+0.178^{*}$ [+0.058, +0.286]	163 vs 115	p=5.9$e-08$
Δ vs. Dense+BM25 RRF	$+0.137^{*}$ [+0.036, +0.215]	$+0.229^{*}$ [+0.074, +0.366]	+0.115 [−0.039, +0.237]	$+0.155^{*}$ [+0.060, +0.246]	163 vs 137	p=0.0022
Δ vs. + Delphi's rerankers	$+0.118^{*}$ [+0.009, +0.198]	$+0.159^{*}$ [+0.007, +0.286]	+0.122 [−0.031, +0.240]	$+0.120^{*}$ [+0.039, +0.203]	163 vs 135	p=0.0011
Δ vs. + expansion	+0.103 [−0.008, +0.189]	+0.087 [−0.100, +0.238]	+0.053 [−0.117, +0.195]	+0.076 [−0.019, +0.157]	163 vs 148	p=0.082
Δ vs. Lexical+BM25 RRF	$+0.172^{*}$ [+0.083, +0.235]	$+0.272^{*}$ [+0.148, +0.372]	$+0.130^{*}$ [+0.024, +0.200]	$+0.172^{*}$ [+0.098, +0.263]	163 vs 131	p=0.00017
Δ vs. Lexical ranker	$+0.188^{*}$ [+0.106, +0.246]	$+0.261^{*}$ [+0.140, +0.361]	$+0.170^{*}$ [+0.088, +0.226]	$+0.158^{*}$ [+0.076, +0.255]	163 vs 123	p=2.4$e-06$
Δ vs. BM25	$+0.204^{*}$ [+0.111, +0.277]	$+0.299^{*}$ [+0.157, +0.433]	$+0.223^{*}$ [+0.111, +0.310]	$+0.221^{*}$ [+0.129, +0.303]	163 vs 113	p=5$e-09$

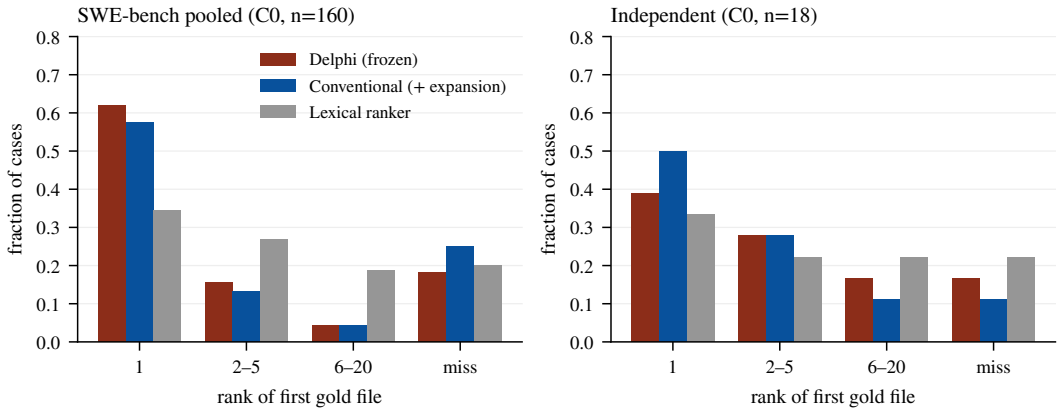

Figure 8: Rank of the first gold file for Delphi, the full conventional stack, and the lexical ranker. Left: 160 pooled SWE-bench instances. Right: the 18-case independent final. "Miss" is no gold file in the top 20.

Exploratory split of the SWE-bench issues. To test the workflow hypothesis outside the C2 set we split the 62 round-3 issues by whether the issue text contains a Python traceback or a path ending in `.py`. Delphi's MRR lead over the full conventional stack was largest on issues with neither (n=35), and smaller on issues with a traceback or a path (n=27). The split was decided after the rankings existed and is reported as exploratory; it does not support the hypothesis that Delphi's structural branches are what separate it on issue-style queries.

Table 9: Pooled SWE-bench Verified (160 C0 instances) by repository. Means of per-instance metrics; repositories ordered by instance count. Repository-cluster intervals in Table 1 resample these twelve clusters.

Repository	n	Delphi		Conventional (+ expansion)		Lexical ranker	
		MRR	R@20	MRR	R@20	MRR	R@20
`django/django`	73	0.691	0.792	0.618	0.696	0.518	0.779
`sympy/sympy`	24	0.729	0.779	0.663	0.739	0.608	0.773
`sphinx-doc/sphinx`	14	0.615	0.643	0.607	0.607	0.227	0.679
`matplotlib/matplotlib`	11	0.682	0.818	0.530	0.727	0.223	0.500
`scikit-learn/scikit-learn`	9	0.815	0.889	0.623	0.722	0.396	0.889
`astropy/astropy`	7	0.548	0.643	0.762	0.786	0.275	0.762
`pydata/xarray`	7	0.929	1.000	0.571	0.500	0.307	0.857
`pytest-dev/pytest`	6	0.708	1.000	0.843	1.000	0.722	0.833
`psf/requests`	3	0.444	0.667	1.000	1.000	0.399	1.000
`pylint-dev/pylint`	3	0.333	0.333	0.083	0.167	0.067	0.167
`mwaskom/seaborn`	2	0.750	1.000	1.000	0.750	0.667	1.000
`pallets/flask`	1	1.000	1.000	1.000	1.000	0.143	1.000

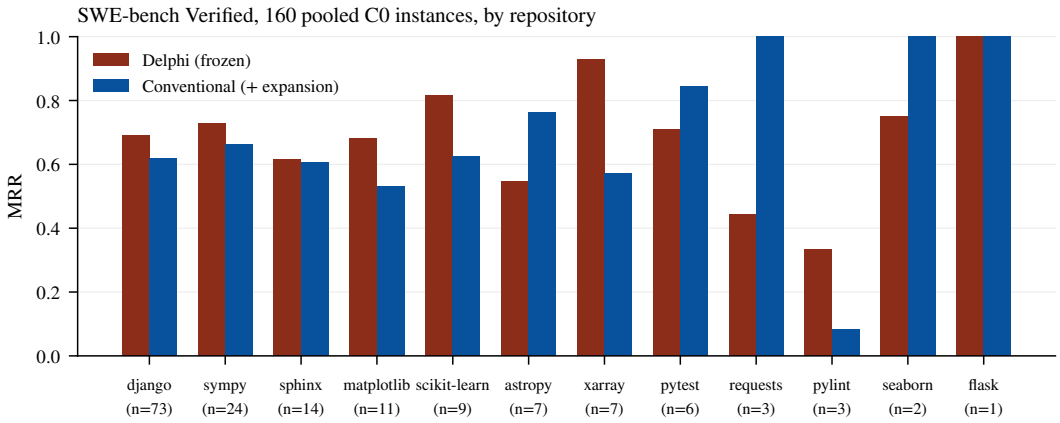

Figure 9: Per-repository MRR on the 160 pooled SWE-bench instances, Delphi against the full conventional stack. Delphi leads on the two largest clusters (`django`, `sympy`) and trails on several small ones; the repository-cluster bootstrap in Table 1 is what turns this heterogeneity into the reported intervals.

Table 10: ARB round-3 partition (C2) by workflow. The concentration of Delphi's lead in `trace2code` and `edit2ripple` is the hypothesis discussed in §4; the partition is validation evidence.

Workflow	n	Delphi		Conventional (+ expansion)		Lexical ranker	
		MRR	R@20	MRR	R@20	MRR	R@20
`trace2code`	38	0.579	0.895	0.149	0.368	0.177	0.711
`edit2ripple`	44	0.383	0.650	0.208	0.642	0.250	0.616
`comment2context`	55	0.303	0.567	0.260	0.691	0.172	0.542
`code2test`	83	0.211	0.588	0.257	0.610	0.055	0.255

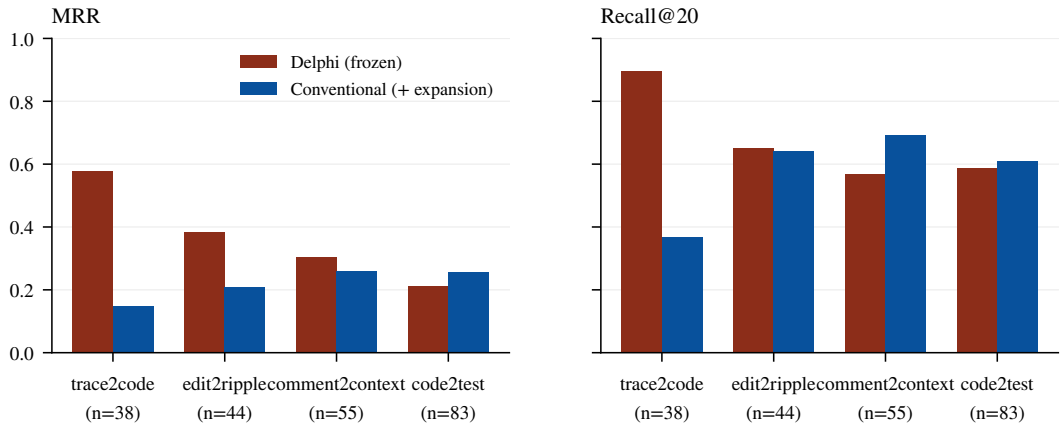

Figure 10: ARB round-3 partition (C2) by workflow, Delphi against the full conventional stack. Queries in `trace2code` carry stack traces with file paths and in `edit2ripple` carry diffs with paths and symbols; `comment2context` and `code2test` carry prose or code without paths.

Table 11: Median query latency in seconds per system and set (wall clock on a warm process, index construction excluded, hosted API round trips included). The lexical comparators' round-3 SWE-bench latencies were recorded in round 3 and are about five times their expansion-set values on the same repositories; the ranker code is identical, so the difference is harness conditions, and we do not compare those two columns.

System	Independent	SWE-bench r3	SWE-bench exp.	ARB (C2)
BM25	1.23	10.66	2.19	1.27
Lexical ranker	1.11	11.03	2.20	1.37
Lexical+BM25 RRF	1.57	13.22	4.19	1.93
Dense	0.49	0.51	0.75	0.50
Dense+BM25 RRF	0.20	2.02	2.04	0.53
+ Delphi's rerankers	2.25	3.72	3.78	2.14
+ expansion	4.06	6.20	6.18	4.22
Delphi (frozen)	3.67	3.63	6.36	4.31

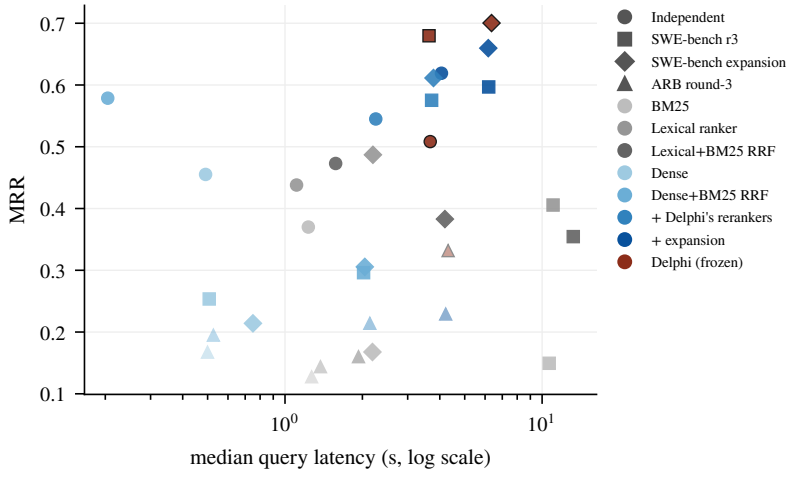

Figure 11: Median query latency against MRR for every system and set (marker shape: set; colour: system; ARB shown faded). The two hosted-LLM rungs and Delphi occupy the same latency band, 2–6 seconds per query; the dense rung is an order of magnitude faster and, on the independent set, competitive.

E DEVELOPMENT ABLATIONS AND NEGATIVE RESULTS

Round-3 preregistered ablations (C1 development sets). Exact vector scan replaced approximate (HNSW) search: $+0.020$ Recall@20 $[0.000, 0.063]$ on the ARB development set, with approximate search additionally implicated in run-to-run variance (Appendix G); accepted. Query expansion off versus on: expansion stayed on. Gemini embedding swap in place of `text-embedding-3-small`: MRR -0.054 $[-0.102, -0.011]$; rejected. File-level Okapi BM25 branch (whole-file BM25 as a seventh fused branch): the single approved configuration failed three preregistered gates (MRR interval floor -0.050 and BCY floor -0.037 against a -0.01 gate on each; warm median latency ratio 1.34 against a 1.20 gate) despite a non-negative Recall@20 delta and zero any-gold losses; rejected and shipped default-off. Two development findings moved code: review-comment queries retrieved the file the comment already quoted, so paths present in the query envelope are demoted after the cross-encoder (quoted-path demotion); and a global exclusion of generated protobuf sources had silently dropped a gold file, so agent-mode indexing now retains generated source under the existing size and binary guards.

Round-2 components, accepted on the ARB v2 pool (now C2). Aligning the query-time embedding model with the index moved workflow-macro MRR from 0.055 to 0.173; reciprocal-rank fusion replaced max-normalized score fusion; the path-affinity branch made file paths searchable; the `ms-marco-MiniLM` cross-encoder at k=30, α=0.4 moved MRR $0.176 \rightarrow 0.193$; hypothetical-document expansion moved held-out MRR $0.228 \rightarrow 0.241$ and Recall@20 $0.552 \rightarrow 0.579$; listwise reranking moved held-out MRR $0.241 \rightarrow 0.285$, Recall@5 $0.355 \rightarrow 0.419$, and BCY@8k $0.376 \rightarrow 0.446$. These are the decisions that make the round-3 ARB partition C2.

Round-2 ideas measured and rejected. Recorded so that the negative space of the design is visible. Four of these looked like wins on the 75-case development split and lost on the 220-case held-out split; development differences under about 0.03 were not trustworthy at that size, and run-to-run variance with an LLM in the loop was about 0.02.

- *File-evidence aggregation* (best chunk plus damped support from other chunks): Recall@20 $0.544 \rightarrow 0.467$ at w=0.5; many gold files match on exactly one chunk.
- *Deeper candidate pools* (50, 100, 150 per branch): MRR $0.197/0.195/0.194$; candidate supply was not the constraint.
- *Deeper rerank window* (k=30, 60, 100): Recall@20 $0.560/0.552/0.444$; the cross-encoder promotes plausible files from the tail.
- *Pure cross-encoder ordering* (α=1.0): MRR $0.193 \rightarrow 0.154$, the worst configuration tested.
- *Larger code-aware reranker* (`bge-reranker-base`, 278M parameters): lost on every metric and ran 7× slower (21.3 s vs. 2.9 s).
- *Fusion weight rebalancing* (five configurations): defaults at or near the optimum; lexical-heavy clearly worse once embeddings were aligned.
- *Reverse-dependency branch* at a global weight: `edit2ripple` Recall@5 $0.238 \rightarrow 0.381$ but overall MRR $0.193 \rightarrow 0.163$; gated on query intent, the harm and the gain both vanished (held-out MRR $+0.004$, Recall@5 -0.007) because the listwise stage already promotes dependents.
- *File path prepended to rerank passages*: development Recall@5 $0.327 \rightarrow 0.333$, held-out $0.355 \rightarrow 0.346$.
- *Wider listwise pool* (20 to 40 candidates): $+0.016$ Recall@20 for -0.016 MRR and -0.008 Recall@5 and $+1.2$ s.
- *Cascade listwise* (second pass over the top 6 with 1,200-character excerpts): $+0.001$ MRR for -0.036 Recall@5 and -0.017 Recall@20.
- *Listwise window and excerpt tuning* (k=12, 700 characters): development Recall@5 0.411 vs. 0.391, held-out 0.378 vs. 0.419.

The diagnostic that these attempts shared: on the held-out split 70.5% of cases had a gold file in the top 20 and 54.1% in the top 5, so the remaining gap was inside the reranker's window, and no change to coverage, fusion, pool size, or prompt shape moved it. The round-4 ladder makes the same point from the other direction: the reranking head is where the margin is, and it is not specific to Delphi.

F DOCUMENTATION TRACK

Cases. Forty DS-1000 problems (development subset, C1), eight each from NumPy, pandas, Matplotlib, scikit-learn, and TensorFlow, each annotated with a gold API identifier that a correct answer must name. Identifier hit is a case-insensitive substring match of the gold identifier against the scored text.

Arms. *Documentation engine* (hosted): library resolution followed by documentation retrieval; two modes were recorded, "oracle" (the library ID chosen by us) and "quality-guided" (the engine's own top-ranked library), both compacted to the 8,000-token budget. *Delphi*: the frozen build indexing the same libraries' documentation sources, k=5 and k=20, compacted to the same budget. *Synthesis engine* (hosted): its native answer mode, recorded once per case with its five cited source chunks; its raw-retrieval mode returned no sources during the recorded round. *Reconstructed synthesis-engine retrieval*: the five source chunks extracted from each recorded response (mean 2,754 tokens per case), packed with the identical routine and budget. *Frozen synthesis stage*: `gpt-5.4-mini`, temperature 0, 1,600 output tokens, the system prompt below, three repeats per arm. *Control*: the same model and repeats with the control prompt and no retrieved context.

Synthesis system prompt (verbatim).

```
You are a documentation-grounded coding assistant.
Answer the coding question directly and concretely.
Name the exact library APIs, functions, methods, and
parameters that solve the task, using their precise
identifiers.
Ground your answer in the retrieved documentation when it is
relevant, and cite the source paths you used.
If the retrieved documentation is missing or unhelpful,
still answer from your own knowledge of the library and say
the documentation did not cover it.
Do not use Markdown code fences.
```

The user turn is `<coding_question>...</coding_question>` followed by `<retrieved_documentation>...</retrieved_documentation>`.

Control system prompt (verbatim).

```
You are a coding assistant.
Answer the coding question directly and concretely.
Name the exact library APIs, functions, methods, and
parameters that solve the task, using their precise
identifiers.
Answer from your own knowledge of the library.
Do not use Markdown code fences.
```

Table 12: Matched-output-contract documentation comparison, 40 development cases (C1) × 3 repeats through one frozen synthesis stage. Deltas are case-cluster bootstrap 95% intervals against the no-retrieval control. The synthesis engine's native single pass is shown for reference only.

Arm	Identifier hit	Per-repeat	Δ vs. control
Documentation engine retrieval + frozen synthesis	0.625	0.650/0.600/0.625	+0.133 [+0.008, +0.258]
Delphi retrieval + frozen synthesis	0.617	0.600/0.625/0.625	+0.125 [+0.000, +0.250]
Synthesis engine retrieval + frozen synthesis	0.583	0.625/0.525/0.600	+0.092 [−0.008, +0.200]
Synthesis engine, native synthesis (recorded)	0.575	single pass	—
Model alone (no retrieval)	0.492	0.475/0.475/0.525	—

Raw retrieval. Identifier hit on the raw retrieved context: Delphi k=5 0.400, k=20 0.425; documentation engine 0.375 in both the quality-guided and the oracle-ID mode; reconstructed synthesis-engine sources 0.475. The synthesis engine's raw retrieval is therefore ahead of Delphi by 0.05–0.075 on this set, a difference the frozen synthesis stage then erases (Table 12).

Table 13: Engine-versus-engine paired differences under the matched contract (case-cluster bootstrap 95% intervals; wins/losses/ties over 40 cases). Every interval includes zero.

Comparison	Δ	95% CI	W/L/T
Delphi − synthesis engine (matched)	$+0.033$	$[-0.092, +0.158]$	7/5/28
Delphi − synthesis engine (native)	$+0.042$	$[-0.092, +0.183]$	6/6/28
Delphi − documentation engine	-0.008	$[-0.117, +0.092]$	6/5/29
Documentation engine − synthesis engine (matched)	$+0.042$	$[-0.083, +0.167]$	10/7/23
Synthesis engine matched − native	$+0.008$	$[-0.050, +0.075]$	5/4/31

DS-1000 execution. A separate arm executes generated code against the DS-1000 tests with a frozen generator (40 cases $\times$ 3 repeats): no retrieval 0.575; documentation engine 0.567; Delphi 0.592; synthesis engine 0.642. Every paired interval against the control spans roughly ± 0.13 (§7).

G Determinism

Probes. Eight ARB development queries repeated ten times each against the pre-freeze stack (mean exact top-10 rate 30.6%), with paired traces of every stage. Three sources were isolated: the listwise and expansion stages were uncached and unseeded; the fusion SQL broke ties by insertion order; and approximate nearest-neighbour search returned different neighbour sets under concurrent load. Fixes: total ordering in every SQL stage (score, then path, then chunk ID), single-flight sharing of concurrent identical hosted calls, an explicit LLM seed, and a persistent stage cache keyed by model, prompt, and seed. After the fixes: 100% exact repeats in-process across two concurrent 8×10 packets and two 75-case repeats; a cold restart moved $7/8$ lists until the cache was warm, after which $8/8$ survived two restarts.

Embedding stability. `text-embedding-3-small`: N=20 repeats of the same texts differed at the fifth decimal and never changed top-ten membership; Gemini embeddings were bitwise identical across repeats.

Like-for-like measures (Table 2, Figure 12). Ten documentation cases $\times$ ten runs per engine, all $\binom{10}{2} \times 10 = 450$ run pairs. *Retrieved set exact*: the set of retrieved units is identical (units: retrieved items for Delphi and the documentation engine; the set of cited URLs for the synthesis engine, whose retrieval is otherwise unobservable). *Retrieved order exact*: the ordered list is identical. *Context byte-exact*: the delivered text is identical. *Identifier-hit agreement*: the scored outcome agrees across the pair. A comparison of Delphi's and the documentation engine's byte-exact context rate with the synthesis engine's byte-exact *answer* rate measures different quantities and is the comparison Table 2 replaces.

H Executable pilot

Protocol (preregistered before any run). Tasks: the 62 round-3 SWE-bench Verified instances (C0). Agent: mini-SWE-agent 2.4.6, text-based action format, `gpt-5.4-mini`, official `x86_64` instance images run under emulation on an Apple-silicon host. Conditions: *no seed*; *random* (five candidate files drawn with seed 1042 per instance, gold excluded); *Delphi* (top five of the frozen build's recorded ranking); *conventional* (top five of the full conventional stack's recorded ranking). Seed block: identical wording, up to five paths with the first 40 lines of each file at the base commit, capped at 3,000 tokens. Budgets: step limit 50 with a 2.0 USD cost cap; a tighter secondary budget was preregistered and not run because agents used 7–8 steps of 50. Outcome: verified repair under the official SWE-bench harness (v5.0.2) against the SWE-bench Verified release; secondary outcomes API cost, steps, and submission rate. Analysis: paired by instance; instance-cluster and repository-cluster bootstrap 95% intervals (20,000 resamples, seed 1042); exact McNemar on discordant pairs. An interval including zero at n=62 is reported as unresolved. Repeats: two independent trajectories per instance and condition (496 trajectories), pooled as per-instance means.

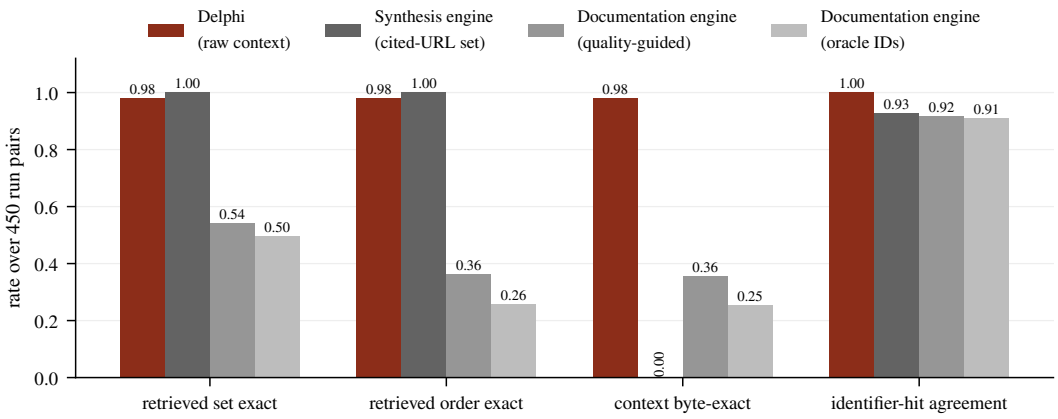

Figure 12: Like-for-like repeatability over 450 run pairs per engine (Table 2). The synthesis engine's cited-source set never changes while its answer text always does; Delphi's retrieved text is byte-exact in 98% of pairs and its scored outcome in 100%.

Seed block (verbatim shape).

```
<retrieved_context>
A code search engine retrieved the following repository
files as potentially relevant to this issue, best match
first.  Treat them as hints and verify them with your own
tools; the fix may involve other files.
### path/to/first_file.py (lines 1-40)
...first 40 lines of the file at the base commit ...
### path/to/second_file.py (lines 1-40)
...
</retrieved_context>
```

When the 3,000-token budget would be exceeded, a file's head is dropped and only its path is listed; when even the path does not fit, the block ends. The random condition uses the same block with five files drawn uniformly from the snapshot's non-gold files.

Table 14: Pooled pilot results: per-instance means over the two repeats. * instance-cluster interval excludes zero.

Condition	Resolved (mean of 2 repeats, of 62)	Rate	Mean cost (USD)	Mean steps
No seed	25.0	0.403	0.0190	8.0
Random files (control)	28.5	0.460	0.0251	8.3
Delphi top-5 (frozen)	28.0	0.452	0.0212	7.2
Conventional ladder top-5	24.0	0.387	0.0232	7.5
Paired differences in per-instance resolved rate: instance-cluster 95% CI [repository-cluster CI]; instances favouring a/b				
Delphi − no seed	$+0.048$ $[-0.008, +0.113]$ $[[+0.017, +0.106]]$; 6/2			
Delphi − random	-0.008 $[-0.073, +0.056]$ $[[-0.081, +0.022]]$; 6/8			
Delphi − conventional	$+0.065$ $[+0.000, +0.137]$ $[[+0.013, +0.147]]$; 12/5			
Conventional − no seed	-0.016 $[-0.089, +0.056]$ $[[-0.106, +0.062]]$; 5/8			
Conventional − random	-0.073^{*} $[-0.145, -0.008]$ $[[-0.186, -0.024]]$; 3/9			
Random − no seed	$+0.056$ $[-0.016, +0.137]$ $[[+0.005, +0.179]]$; 9/5			

Reading the repeats. In the first repeat the Delphi seed resolved 31 instances against 25 for no seed, an apparent $+0.097$ that a single-repeat analysis would have reported with an instance-cluster interval excluding zero; in the second repeat both resolved 25. The conventional seed resolved 23 in the first repeat and 25 in the second. Pooled over repeats, the only interval excluding zero is the conventional seed's deficit against the random control. Sampling variance across trajectories of the same instance is therefore of the same order as the between-condition differences, and any single-trajectory pilot of this size should be read as inconclusive regardless of its point estimate.

Table 15: First repeat alone.

Condition	Resolved (of 62)	Rate	Mean cost (USD)	Mean steps
No seed	25.0	0.403	0.0162	7.7
Random files (control)	30.0	0.484	0.0263	9.0
Delphi top-5 (frozen)	31.0	0.500	0.0208	7.3
Conventional ladder top-5	23.0	0.371	0.0240	7.5
Paired differences in per-instance resolved rate: instance-cluster 95% CI [repository-cluster CI]; instances favouring a/b				
Delphi − no seed	$+0.097^{*}$ [+0.016, +0.194] [[+0.019, +0.244]]; 7/1			
Delphi − random	+0.016 [−0.065, +0.097] [[−0.057, +0.154]]; 4/3			
Delphi − conventional	$+0.129^{*}$ [+0.032, +0.242] [[+0.029, +0.317]]; 10/2			
Conventional − no seed	−0.032 [−0.113, +0.048] [[−0.147, +0.045]]; 2/4			
Conventional − random	-0.113^{*} [−0.210, −0.032] [[−0.208, −0.059]]; 1/8			
Random − no seed	+0.081 [−0.016, +0.177] [[−0.015, +0.200]]; 7/2			

Table 16: Second repeat alone.

Condition	Resolved (of 62)	Rate	Mean cost (USD)	Mean steps
No seed	25.0	0.403	0.0219	8.3
Random files (control)	27.0	0.435	0.0238	7.6
Delphi top-5 (frozen)	25.0	0.403	0.0216	7.2
Conventional ladder top-5	25.0	0.403	0.0224	7.5
Paired differences in per-instance resolved rate: instance-cluster 95% CI [repository-cluster CI]; instances favouring a/b				
Delphi − no seed	+0.000 [−0.097, +0.097] [[−0.097, +0.061]]; 5/5			
Delphi − random	−0.032 [−0.129, +0.065] [[−0.219, +0.040]]; 4/6			
Delphi − conventional	+0.000 [−0.097, +0.097] [[−0.074, +0.027]]; 4/4			
Conventional − no seed	+0.000 [−0.097, +0.097] [[−0.082, +0.094]]; 5/5			
Conventional − random	−0.032 [−0.129, +0.048] [[−0.189, +0.023]]; 3/5			
Random − no seed	+0.032 [−0.065, +0.129] [[−0.034, +0.194]]; 6/4			

Seed quality. Delphi's top five named at least one gold file on 47 of 62 instances; the full conventional stack's on 42; the random block by construction on none. Agents in every condition retained ordinary shell tools and could, and did, locate files the seed had not named.

Cost and steps. Mean cost per trajectory: no seed \$0.019, random \$0.025, Delphi \$0.021, conventional \$0.023; mean steps 8.0, 8.3, 7.2, 7.5 respectively. Total spend for all 496 trajectories was \$10.97. All but two trajectories terminated by submission; the two exceptions (random condition, first repeat) hit the step or cost limit and count as unresolved.

I HOSTED ENGINE ACCOUNTING

Documentation engine. Accessed through its public REST API with a project credential. Two retrieval modes were recorded (oracle library ID and quality-guided); the quality-guided mode is the fair comparison because it does not receive the gold library. Its retrieved documentation drifted across runs (54% retrieved-set exact), which is the engine's behaviour and not an artifact of our client.

Synthesis engine, documentation surface. Accessed through its hosted API. The native answer mode was recorded once per case in round 3 with its cited sources; the raw-retrieval mode returned no sources during that round. Round 4 reconstructed the retrieved context from the recorded responses rather than issuing new calls, so the matched arm reflects the engine's round-3 retrieval exactly. Determinism probes (ten cases $\times$ ten runs) were issued in round 3.

Synthesis engine, repository surface. Indexing the ARB development split requires 68 exact (repository, base commit) snapshots in the provider's account. Round 2 indexed 32 pairs through a sharded upload; whole-snapshot ingestion of the large repositories entered a "syncing" state that never reached a terminal status. Round 3 issued a minimal one-shard probe that remained syncing across a 40-minute deadline and a three-hour watch. Round 4 issued a fresh one-shard probe under a newly provisioned credential (which the account inventory shows resolves to the same account, with

the earlier probes still syncing) that again timed out after 40 minutes. No repository query was ever scored against the engine, so no number is reported in either direction. We do not attribute the stall to the engine's design; it may be an account, quota, or ingestion-path issue that a different account would not encounter.

Credentials. Hosted runs used project credentials supplied by the authors; none of the recorded artifacts contain them, and re-running the hosted arms requires the reader's own.

J Compute and cost

All round-4 experiments ran on one Apple-silicon workstation with a local PostgreSQL instance, using hosted APIs for embeddings and LLM stages. The matched ladder embedded 233.5M new tokens across its runs (3,807 embedding requests, roughly $4.70 at list price); its persistent embedding cache holds $1{,}440{,}990$ vectors (11.9 GB), and its LLM stage cache holds 1,051 listwise and expansion replies, so any rung can be re-executed bitwise without hosted calls. Delphi's own stage cache for the frozen build is archived separately. The executable pilot ran 496 trajectories (4 conditions $\times$ 2 repeats $\times$ 62 instances) with four concurrent containerized workers under `amd64` emulation for a total API spend of $10.97; harness evaluation ran the official SWE-bench images once per trajectory. The documentation matched arm reused recorded hosted responses and issued 120 frozen-synthesis calls. Round-3 costs (the Delphi runs, lexical comparators, determinism probes, and hosted documentation calls) are recorded in the archive's journals.

K Artifact release

Everything below is contained in a single archive and will be released publicly, together with the frozen Delphi source at the recorded revision, once the de-identification pass is complete; an anonymized copy accompanies the submission and the public URL is withheld for review.

- *Per-case results* for every system on every set, including development and rejected configurations: one record per case with the query, gold paths, ranked paths, per-case metrics, latency, and the served configuration, together with per-run summaries (487 files, 221 MB).
- *Paired analyses*: every bootstrap interval and McNemar count reported in the paper, the like-for-like determinism analysis, and the documentation synthesis outputs for every arm and repeat.
- *The exposure ledger* with every case ID and the evidence for its class, and the expansion index audit.
- *Agent trajectories*: the four seeded datasets, all 496 mini-SWE-agent trajectories with every model call and action, the official harness evaluation reports and logs per condition and repeat, and the pooled and per-repeat analyses.
- *Split manifests and case files* for every set, including the expansion draw manifest with the two preprocessing rules applied.
- *Implementation*: the evaluation runner for every static system (Delphi, the matched ladder, the lexical comparators), the paired-analysis code, the exposure-ledger builder, the corpus restorer (bare clones at recorded commits), the expansion sampler, the pilot dataset builder and analyzer, the documentation-track runners with the synthesis stage and the synthesis-engine context reconstruction, and the scripts that regenerate every table and figure in this paper.
- *Serving*: the frozen Delphi source, the native launch configuration (Appendix A), the pilot run configuration, and the configurations of the round-3 ablations.
- *Journals*: the state, decision, hypothesis, and workstream records; the preregistered pilot protocol; an artifact inventory with SHA-256 of every table's inputs; the round-2 protocol, source and statistics locks, capability audits of the hosted engines, and the measured-and-rejected record (Appendix E).

Repository corpora (58 repositories, 365 snapshots) are not redistributed; a provided script re-clones them from public GitHub at the recorded commits. Hosted-engine artifacts are the recorded responses; re-running those arms needs the reader's own credentials.